ACHAL NAWAL

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PROFESSIONAL SUMMARY

Computer Engineering student with hands-on experience building GenAI/LLM-powered applications, RAG pipelines, and automated data-processing workflows on cloud platforms (Azure OpenAI). Skilled in Python, SQL, and FastAPI for building scalable pipelines, and experienced in translating unstructured data into structured, actionable insights.

EDUCATION

Thapar Institute of Engineering and Technology, Patiala, India
Bachelor of Engineering in Computer Engineering | CGPA: 8.6/10 | 2023 – 2027

Maharaja Sumer Singh Public School, Kishangarh, Rajasthan
CBSE Class XII – 94.4% | 2023

EXPERIENCE

Artificial Intelligence Applications Development Intern
AmberFlux EdgeAI Pvt. Ltd. | May 2026 – Jul 2026
Tech Stack: Python | FastAPI | Azure OpenAI | Azure Document Intelligence | Azure Service Bus | NetworkX | RapidFuzz

- Developed AI-powered document extraction workflows using Azure Document Intelligence and GPT-based pipelines for automated engineering document processing.
- Built components of a concurrent, multi-threaded data pipeline to process PDFs, emails, and technical drawings at scale, applying business rules and data-quality checks across reusable pipeline layers.
- Implemented FastAPI backend services and Azure Service Bus workflows for end-to-end AI pipeline orchestration.
- Built a graph- and fuzzy-matching–based automation system (NetworkX, RapidFuzz) combined with LLM-based classification to reduce manual reconciliation in project- and email-matching workflows.

AI/ML & Cybersecurity Research Intern
University of Queensland | July 2025 – December 2025

- Implemented adversarial training for the HAMMER multimodal deepfake detection model using a CNN-based generator to create image perturbations.
- Improved model robustness by training on mixed clean and adversarial samples and evaluating performance using AUC, EER, and IoU metrics.
- Co-authored "Adv-HAMMER: Adversarially Robust Multimodal Fake Detection with Perturbation-Aware Training," accepted at PP-MisDet @ CVPR 2026.

TECHNICAL SKILLS

Languages: Python, C++, C, JavaScript, HTML, CSS, SQL
GenAI & AI/ML: LLM Application Development, RAG Pipelines, Prompt-based Automation, PyTorch, Scikit-learn, OpenCV, MediaPipe, YOLOv8
Cloud & AI Platforms: Azure OpenAI, Azure Document Intelligence, Azure Service Bus, Google Gemini API
Data & Analytics: PostgreSQL, MySQL, ChromaDB, Data Pipeline Automation, Dashboarding (Streamlit)
Networking & Systems: WebSockets, REST APIs, Windows UIAutomation
Frameworks & Tools: FastAPI, Flask, PyQt6, Docling, NetworkX, RapidFuzz, Git, GitHub, Postman, Docker

PROJECT EXPERIENCE

ChatFusion – Multi-source RAG Knowledge Workspace | GitHub
(Python, Flask, Retrieval Augmented Generation (RAG), ChromaDB, Sentence Transformers, Whisper, Google Gemini API, HTML, CSS, JavaScript)

- Built a multi-modal RAG system enabling conversational querying across YouTube videos, PDF documents, and audio files.
- Implemented cross-document and cross-modal retrieval with synthesis, allowing unified reasoning over multiple data sources.
- Designed an end-to-end retrieval pipeline using Sentence Transformers and ChromaDB for efficient semantic search.
- Developed a Flask backend with REST APIs and an interactive frontend supporting document- and workspace-based querying.

FloatChat – ARGO Ocean Intelligence System | GitHub
(Python, FastAPI, Streamlit, PostgreSQL, ChromaDB, Google Gemini API, NetCDF)

- Built an AI-powered conversational system for querying ARGO oceanographic float data using natural language.
- Designed an automated pipeline to download, ingest, and store ARGO NetCDF files into PostgreSQL, using SQL for structured querying and validation.
- Developed a FastAPI backend and Streamlit dashboards for interactive querying, visualization, and data exploration.

TITAN 2.0 – Multimodal AI Desktop Automation | GitHub
(Python, Gemini Live API, WebSockets, PyQt6, OpenCV, Docling, UIAutomation)

- Engineered a real-time multimodal AI assistant using Gemini Live and full-duplex WebSockets, integrating voice interaction, visual reasoning, tool calling, and interruptible command execution.
- Engineered a bidirectional WebSocket bridge between the desktop AI runtime and Chrome extension, enabling asynchronous browser task execution with command correlation, response handling, and timeout recovery.
- Built an autonomous skill-execution framework that generates and manages Python tools, applying AST-based security checks and isolated subprocess testing with execution timeouts before running generated code.
- Implemented AI security guardrails for prompt-injection, task-override, sensitive-data, and dangerous-action detection, with sanitization, strict-mode validation, and automated tests.

PUBLICATIONS & CERTIFICATIONS

- Adv-HAMMER: Adversarially Robust Multimodal Fake Detection with Perturbation-Aware Training — *PP-MisDet @ CVPR 2026, Accepted*
- Machine Learning with Python - Coursera
- Data Structures and Algorithms in C++ - Udemy
- Introduction to Cybersecurity -Cisco Networking Academy